Oscillations

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Simple Harmonic Motion (SHM)

Definition:

- Any motion that repeats at regular intervals is called *periodic motion* or *harmonic motion*.
- Simple harmonic motion (SHM) is a particular type of periodic motion in which the motion is a *sinusoidal function* of time t. That is, it can be written as a sine or a cosine of time t. For example,

$$x(t) = x_m \sin(\omega t + \phi)$$

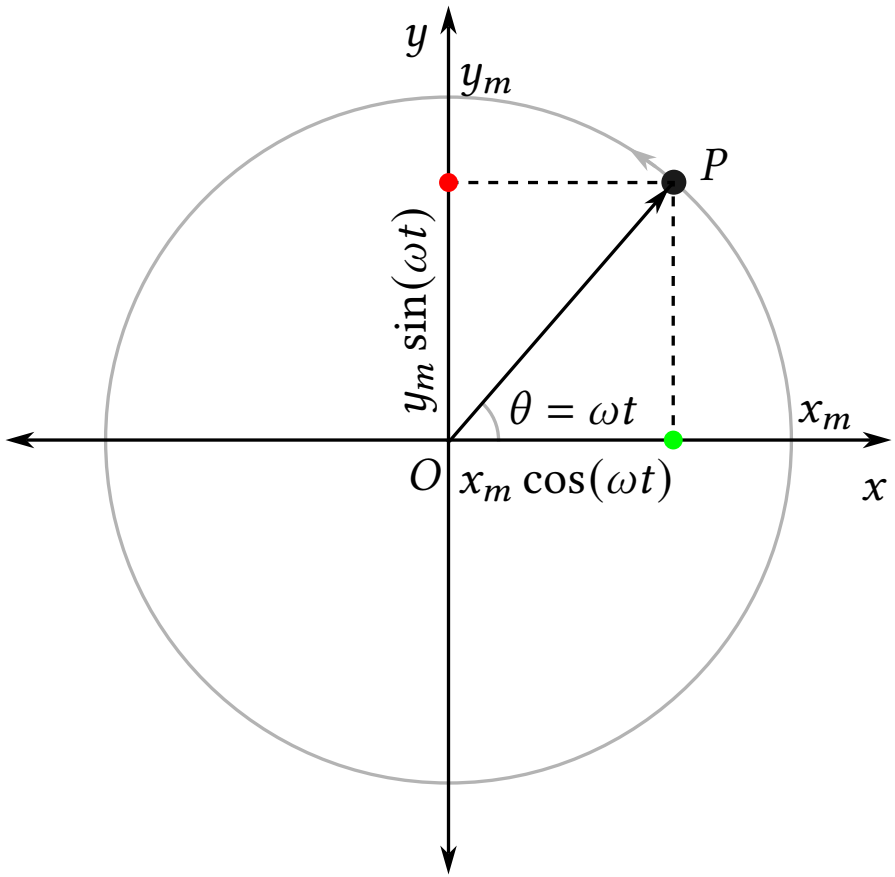
- In simple harmonic motion, the acceleration a is proportional to the displacement x but opposite in sign, and the two quantities are related by the square of the angular frequency ω.

$$a \propto -x$$

- Simple harmonic motion is the motion of a particle when the force acting on it is proportional to the particle's displacement but in the opposite direction.

$$F \propto -x$$

- Linear simple harmonic motion is the projection of *uniform circular motion* on a diameter of the circle in which the circular motion occurs.



Parameters of an Oscillatory System

- **Mean position:** The position of the oscillating body at rest.
- **Amplitude:** The amplitude of an SHM is the maximum displacement of the body from its mean position.
- **Time Period:** The time interval during which the oscillation repeats itself is called the time period. It is denoted by T and its unit is seconds.

$$T = 2\pi\sqrt{\frac{\text{displacement}}{\text{acceleration}}}$$

- **Frequency:** The number of oscillations that a body completes in one second is called the frequency of periodic motion. It is the reciprocal of the time period T and it is denoted by f

$$f = \frac{1}{T}$$

- **Angular frequency:** This is the orbital frequency or circular frequency and it is measured in radians per second. It is denoted by ω.

$$\omega = 2\pi f = \frac{2\pi}{T}$$

- **Phase:** It is the state of an oscillating system. If the SHM is represented by $y = A\ \sin(\omega t + \phi)$, together with ω as the angular frequency. The quantity $(\omega t + \phi)$ of the sine function is called the total phase of the motion at time t and ϕ is the initial phase or epoch.

Types of Simple Harmonic Motion

1. **Linear simple harmonic motion:** If the body executing SHM has a linear acceleration then the motion of the body is linear simple harmonic motion. Linear displacement, x or y is given by

$$x(t) = x_m \sin(\omega t + \phi) \qquad \text{Or} \qquad y(t) = y_m \sin(\omega t + \phi)$$

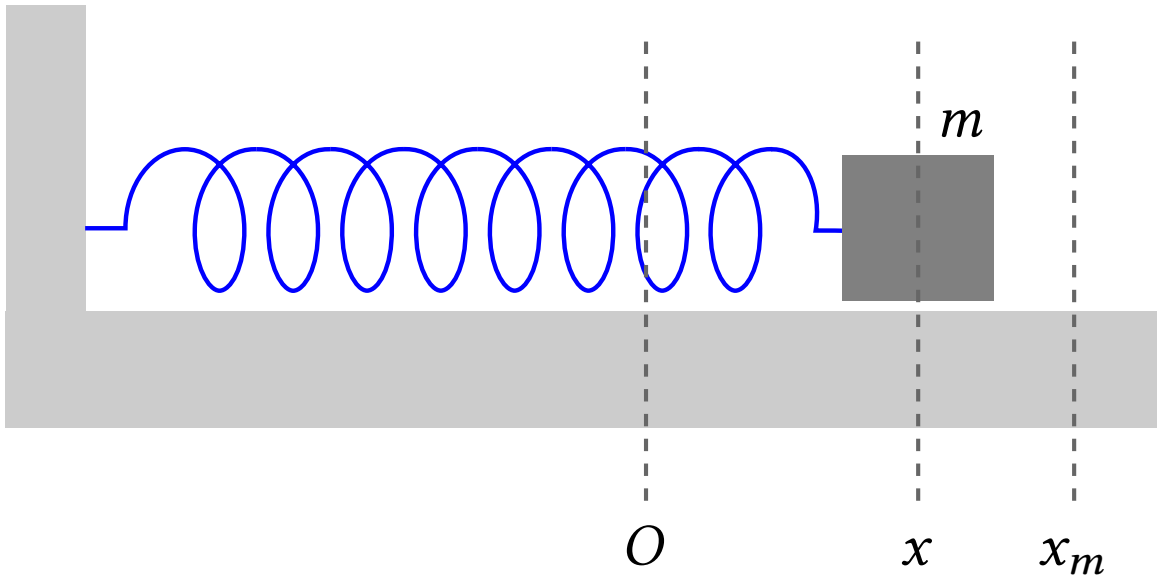
   Examples: spring and mass system.

2. **Angular simple harmonic motion:** If the body executing SHM has an angular acceleration then the motion of the body is angular simple harmonic motion. Angular displacement, θ is given by

$$\theta(t) = \theta_m \sin(\omega t + \phi)$$

   Examples: simple pendulum, torsional pendulum.

Spring and Mass System



Restoring force in the spring is proportional to the stretch. Hence, this force can be written as,

$$F = -kx$$

where k is called the spring constant. From Newton's 2nd law,

$$F = m\frac{dv}{dt}$$

From these two equations,

$$m\frac{dv}{dt} = -kx$$

Writing

$$\frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = \frac{dv}{dx}v$$

we get

$$m\frac{dv}{dx}v = -kx$$

Rearranging and integrating on both sides,

$$m\int_0^v v\,dv = -k\int_{x_m}^x x\,dx$$

$$\frac{1}{2}mv^2 = \frac{k}{2}(x_m^2 - x^2)$$

$$\implies v = \sqrt{\frac{k}{m}}\sqrt{x_m^2 - x^2}$$

Writing $v = dx/dt$, rearranging and integrating on both sides,

$$\int_{x_m}^{x} \frac{dx}{\sqrt{x_m^2 - x^2}} = \sqrt{\frac{k}{m}} \int_0^t dt$$

$$\left[\arcsin\left(\frac{x}{x_m}\right)\right]_{x_m}^{x} = \sqrt{\frac{k}{m}} t$$

$$\arcsin\left(\frac{x}{x_m}\right) - \arcsin(1) = \sqrt{\frac{k}{m}} t$$

$$\arcsin\left(\frac{x}{x_m}\right) = \sqrt{\frac{k}{m}} t + \frac{\pi}{2}$$

$$x = x_m \sin\left(\sqrt{\frac{k}{m}} t + \frac{\pi}{2}\right) = x_m \cos\left(\sqrt{\frac{k}{m}} t\right)$$

Here $\sqrt{\frac{k}{m}} = \omega$ is the angular frequency and x_m is the amplitude (maximum displacement).

$$\therefore \quad \boxed{x = x_m \cos(\omega t)}$$

Time Period:

For one complete oscillation, which takes time T, the angle spanned by the cosine or sine functions is 2π. Therefore,

$$\omega T = 2\pi$$

where $\omega = \sqrt{\frac{k}{m}}$.

$$\therefore \quad T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$$

Frequency is,

$$f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

Another way:

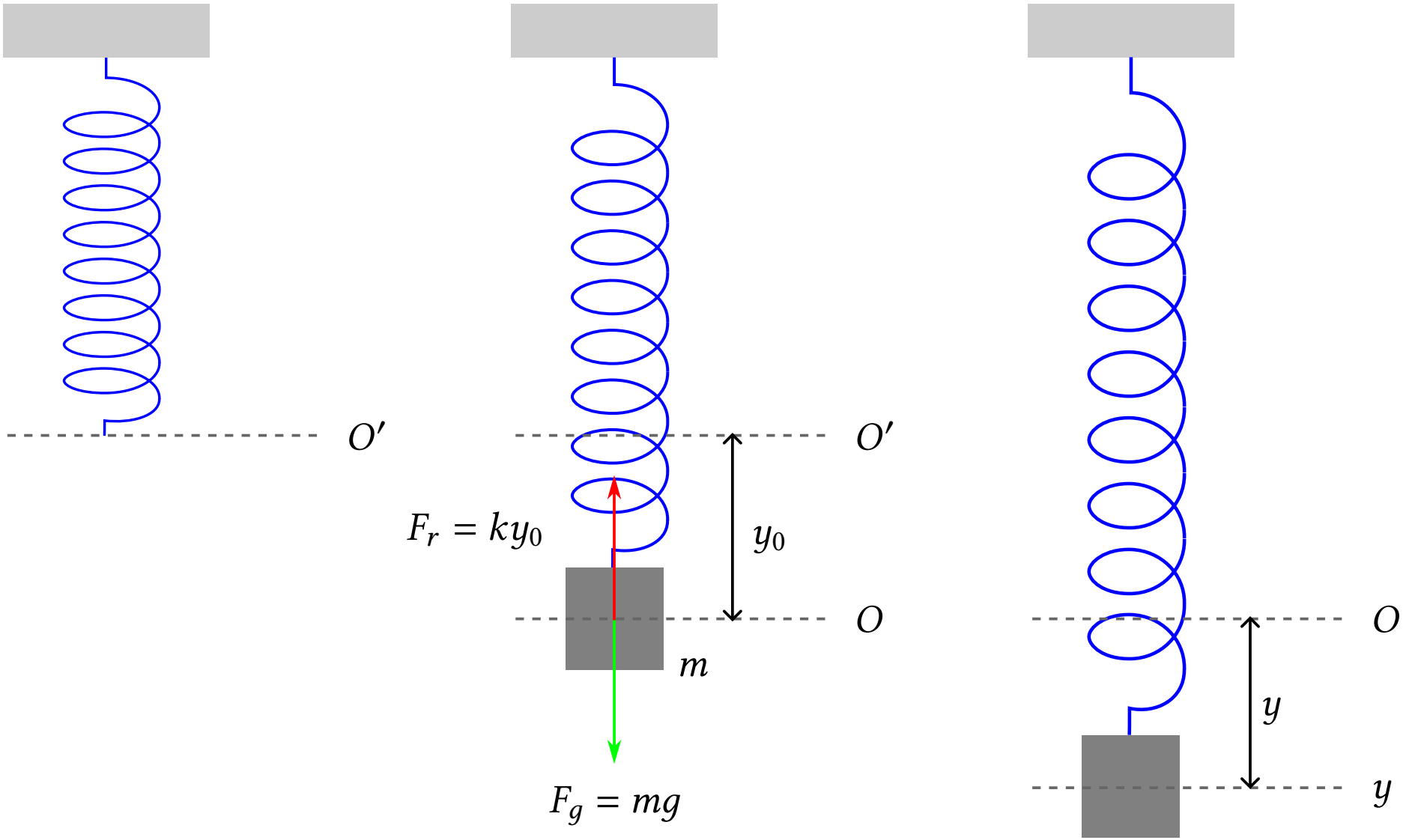
$$\text{displacement} = x$$

From $m\frac{dv}{dt} = -kx$

$$\text{acceleration} = -\frac{k}{m}x$$

$$\therefore \quad T = 2\pi\sqrt{\frac{\text{displacement}}{\text{acceleration}}} = 2\pi\sqrt{\frac{x}{kx/m}} = 2\pi\sqrt{\frac{m}{k}}$$

Vertical Spring and Mass System under Gravity



After application of load to the spring it stretches by y_0. Now the force balance equation is,

$$ky_0 = mg$$

By rearranging,

$$y_0 = \frac{mg}{k} \qquad \text{or} \qquad k = \frac{mg}{y_0}$$

It is a way to know the spring constant k.

Here constant force due to gravity, which is equal to mg, is always counter balanced by the restoring force developed in the spring. Hence, the spring oscillates with this new position O as the equilibrium or mean position.

Total force on the spring at any time will be,

$$\begin{aligned} F &= F_{\text{restore}} + mg \\ &= -k(y + y_0) + mg \\ &= -ky - ky_0 + mg \\ &= -ky - \cancel{mg} + \cancel{mg} \\ &= -ky \end{aligned}$$

Hence Newton's second law reads,

$$m\frac{d^2y}{dt^2} = -ky$$

$$\frac{d^2y}{dt^2} = -\frac{k}{m}y$$

$$\frac{d^2y}{dt^2} = -\omega^2 y$$

where $\omega = \sqrt{\frac{k}{m}}$.

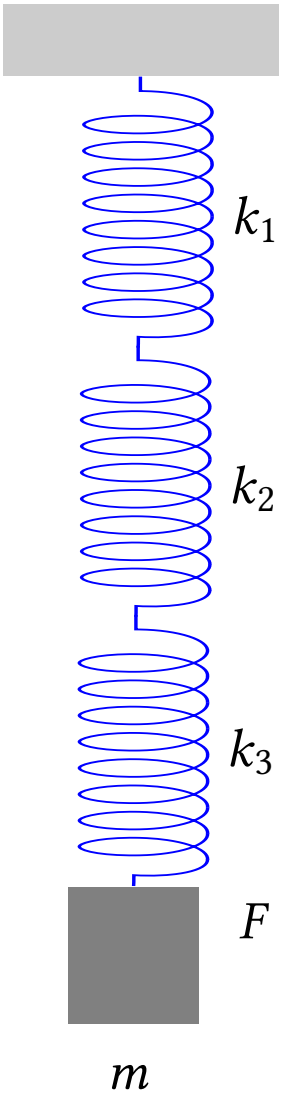
By inspection, its solution is,

$$y = y_m \cos(\omega t + \phi) \qquad \text{or} \qquad y = y_m \sin(\omega t + \phi)$$

Verification:

$$\begin{aligned}
\text{LHS} = \frac{d^2y}{dt^2} &= \frac{d^2}{dt^2}[y_m \cos(\omega t + \phi)] \\
&= \frac{d}{dt}[-y_m \omega \sin(\omega t + \phi)] \\
&= -y_m \omega^2 \cos(\omega t + \phi) \\
&= -\omega^2 y = \text{RHS}
\end{aligned}$$

Springs in Series



If y is the total extension in the springs and y_1, y_2, y_3 that of springs in series, then

$$y = y_1 + y_2 + y_3$$

Since all the springs are subjected to same force mg,

$$F = -k_1 y_1 = -k_2 y_2 = -k_3 y_3$$

$$\implies \quad y_1 = -\frac{F}{k_1},\ y_2 = -\frac{F}{k_2},\ y_3 = -\frac{F}{k_3}$$

Putting these into first equation,

$$y = -\frac{F}{k_1} - \frac{F}{k_2} - \frac{F}{k_3} = -F\left(\frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}\right)$$

With

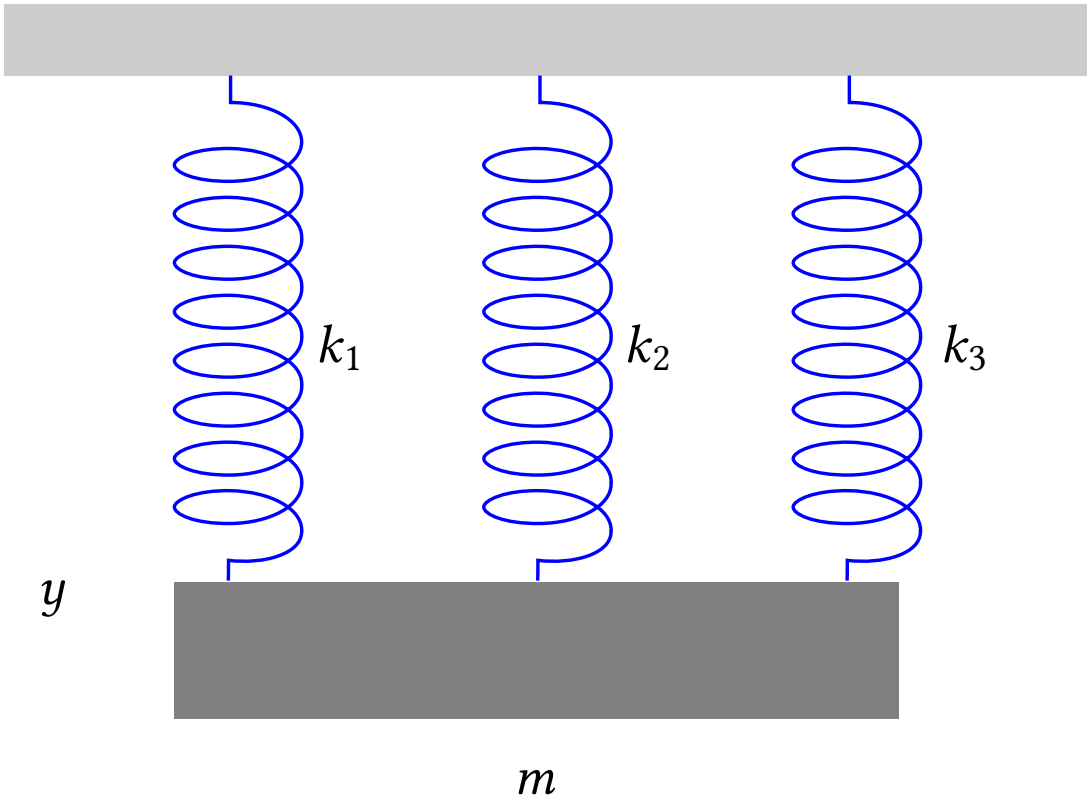
$$F = -k_{\text{eff}} y$$

$$\implies y = -F \frac{1}{k_{\text{eff}}}$$

Comparing these two,

$$\frac{1}{k_{\text{eff}}} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}$$

Springs in Parallel



If F is the total force on the springs and F_1, F_2, F_3 that of springs in series, then

$$F = F_1 + F_2 + F_3$$

Since all the springs experience same stretch y,

$$F_1 = -k_1 y,\ F_2 = -k_2 y,\ F_3 = -k_3 y$$

Putting these into first equation,

$$F = -k_1 y - k_2 y - k_3 y = -(k_1 + k_2 + k_3) y$$

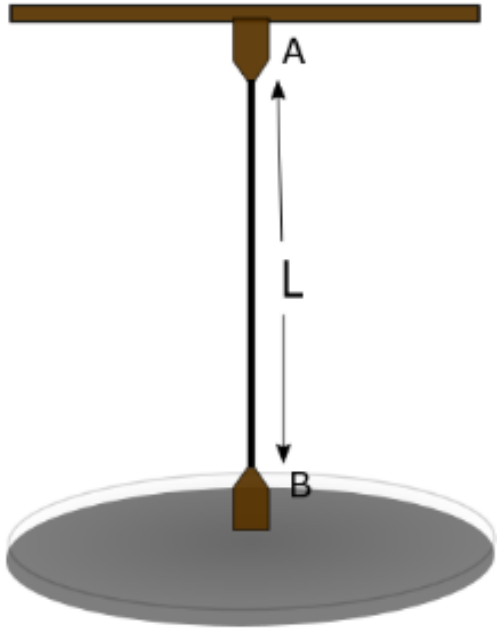
With

$$F = -k_{\text{eff}} y$$

Comparing these two,

$$k_{\text{eff}} = k_1 + k_2 + k_3$$

Torsion Pendulum



- The disc executes torsional oscillations about the mean angular position.
- If I is the moment of inertia of the disc about the axis of the wire and $d^2\theta/dt^2 = \alpha$ is the angular acceleration, then

$$\text{deflecting couple} = I\alpha$$

- If θ is the angle of twist in the wire and C is the couple per unit twist for the wire, then

$$\text{restoring couple} = -C\theta$$

The negative sign indicates that the restoring couple is in the opposite direction to the deflecting couple.

- At any instant:

$$\text{deflecting couple} = \text{restoring couple}$$

$$\therefore \quad I\alpha = -C\theta$$

$$\frac{d^2\theta}{dt^2} + \frac{C}{I}\theta = 0$$

Calling $C/I = \omega^2$,

$$\frac{d^2\theta}{dt^2} + \omega^2\theta = 0$$

- This is the differential equation for angular simple harmonic motion whose solution is,

$$\therefore \quad \boxed{\theta = \theta_m \cos(\omega t)}$$

with $\theta = \theta_m$ at $t = 0$.

Time Period:

For one complete oscillation, which takes time T, the angle spanned by the cosine or sine functions is 2π. Therefore,

$$\omega T = 2\pi$$

where $\omega = \sqrt{\frac{C}{I}}$.

$$\therefore \quad T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I}{C}}$$

Frequency is,

$$f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{C}{I}}$$

Another way:

$$\text{angular displacement} = \theta$$

From $I\frac{d^2\theta}{dt^2} = -C\theta$

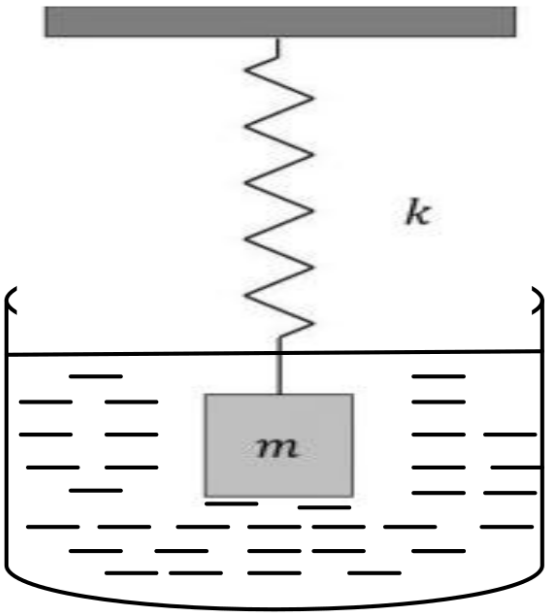
$$\text{angular acceleration} = -\frac{C}{I}\theta$$

$$\therefore \quad T = 2\pi\sqrt{\frac{\text{displacement}}{\text{acceleration}}} = 2\pi\sqrt{\frac{\theta}{C\theta/I}} = 2\pi\sqrt{\frac{I}{C}}$$

Note: For a given wire of length l, radius r and rigidity modulus η,

$$C = \frac{\pi\eta r^4}{2l}$$

Damped Harmonic Oscillator



- A block of mass m suspended from a fixed end by means of a mass less spring in a viscous medium.
- If it is set into vibrations then its amplitude gradually decreases with time.
- As the mass vibrates in the liquid there will be an opposing viscous force developed.
- The viscous force which opposes the motion is proportional to the relative speed between the mass and the liquid.

If y is the displacement of the body from the equilibrium state at any instant of time t, then dy/dt is the instantaneous velocity and d^2y/dt^2 is the acceleration.

At any instant the two forces acting on the body are:

a) A restoring force which is proportional to displacement and acts in the opposite direction:

$$F_{\text{restoring}} = -ky$$

where k is the spring constant.

b) A frictional or damping force which is directly proportional to the velocity of the mass and is opposite to the motion:

$$F_{\text{damping}} = -r\frac{dy}{dt}$$

where r is the damping constant.

The net force acting on the oscillator:

$$F_{\text{restoring}} + F_{\text{damping}} = -ky - r\frac{dy}{dt}$$

By Newton's 2nd law of motion:

$$F = m\frac{d^2y}{dt^2} = -ky - r\frac{dy}{dt}$$

$$\text{or} \qquad \frac{d^2y}{dt^2} + \frac{r}{m}\frac{dy}{dt} + \frac{k}{m}y = 0 \tag{1}$$

Let $r/m = 2b$ and $k/m = \omega^2$

$$\frac{d^2y}{dt^2} + 2b\frac{dy}{dt} + \omega^2 y = 0 \tag{2}$$

This is a differential equation of second order its solution will be in the form of,

$$y = Ae^{\alpha t} \tag{3}$$

Where the constants A and α are to be found.

To get the constants we differentiate (3),

$$\frac{dy}{dt} = A\alpha e^{\alpha t} \qquad \text{and} \qquad \frac{d^2y}{dt^2} = A\alpha^2 e^{\alpha t}$$

By substituting these in (2),

$$A\alpha^2 e^{\alpha t} + 2bA\alpha e^{\alpha t} + \omega^2 A e^{\alpha t} = 0$$

$$\text{or} \qquad A e^{\alpha t}(\alpha^2 + 2b\alpha + \omega^2) = 0$$

For this either $y = Ae^{\alpha t} = 0$ (trivial solution) or

$$(\alpha^2 + 2b\alpha + \omega^2) = 0$$

$$\therefore \quad \alpha = \frac{-2b \pm \sqrt{4b^2 - 4\omega^2}}{2} = -b \pm \sqrt{b^2 - \omega^2}$$

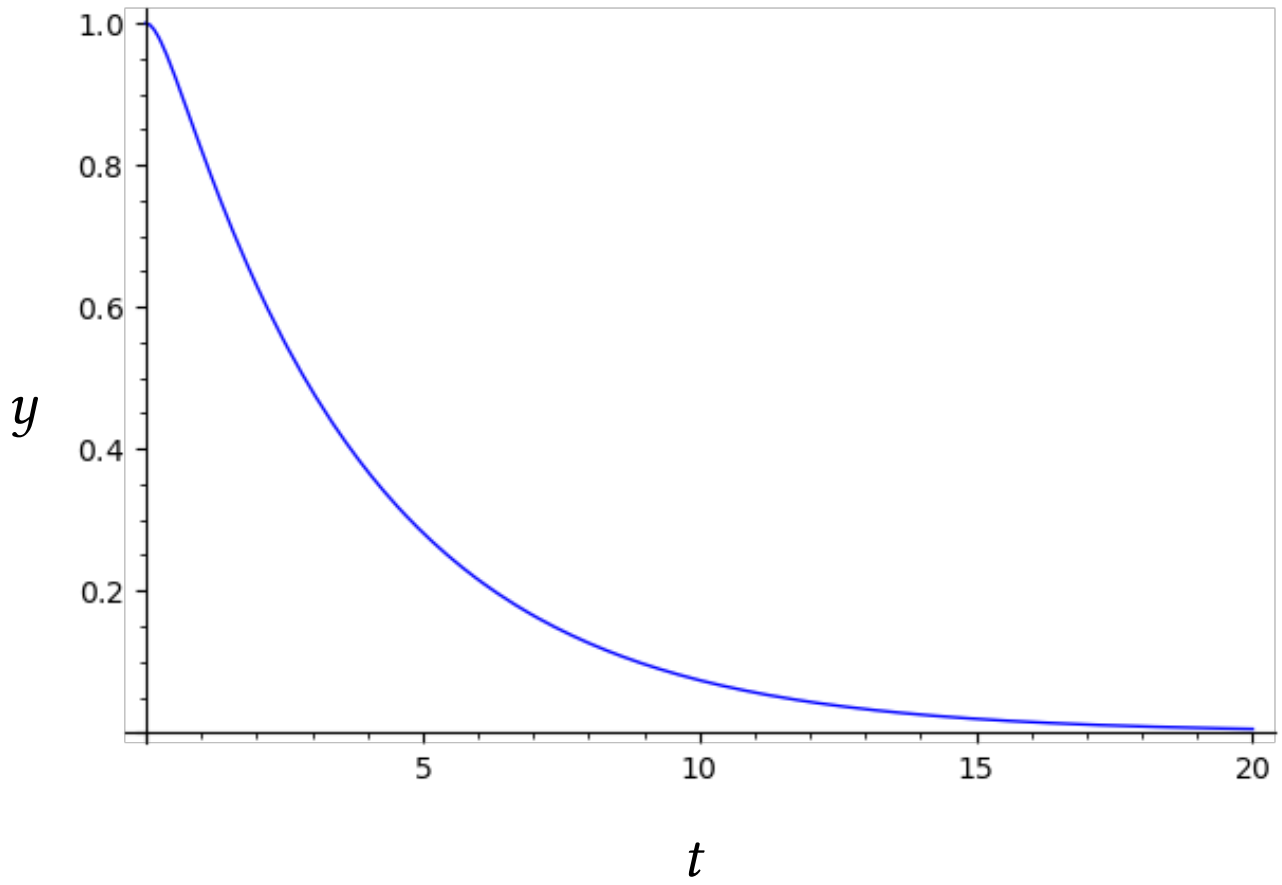
Hence the general solution of (2) is,

$$\boxed{y = Ce^{\left(-b+\sqrt{b^2-\omega^2}\right)t} + De^{\left(-b-\sqrt{b^2-\omega^2}\right)t}} \tag{4}$$

Where C and D are integral constants. The actual solution depends upon whether $b^2 > \omega^2$, $b^2 = \omega^2$ or $b^2 < \omega^2$.

Case 1: Heavy damping or over damping ($b^2 > \omega^2$)

- In this case, $\sqrt{b^2 - \omega^2}$ is real and less than b, therefore in equation (4) both the exponents are negative.
- It means that the displacement y of the particle decreases continuously with time.
- That is, the particle when once displaced returns to its equilibrium position slowly without performing any oscillations.
- Such a motion is called *overdamped* or *aperiodic* motion.
- This type of motion is shown by a pendulum moving in thick oil or by a deadbeat moving coil galvanometer.


1.0
0.8
0.6
0.4
0.2
y
5
10
15
20
t

Case 2: Critical damping ($b^2 = \omega^2$)

By substituting $b^2 = \omega^2$ in (4) results in,

$$y = Ce^{-bt} + De^{-bt} = (C + D)e^{-bt} = Ee^{-bt}$$

This is an incomplete solution because mathematically solution of a second order linear differential equation must always involve two arbitrary constants of integration. Physically two constants specify initial position and initial velocity.

For this instead consider $b^2 \approx \omega^2$ so that $\sqrt{b^2 - \omega^2}$ is very close to 0 but not equal to zero, a limit tending to 0. Call $\sqrt{b^2 - \omega^2} = \beta$ and write the solution as,

$$\begin{aligned} y &= Ce^{(-b+\beta)t} + De^{(-b-\beta)t} \\ &= e^{-bt}\left[Ce^{\beta t} + De^{-\beta t}\right] \end{aligned}$$

We use the series expansion

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

and write

$$e^{\beta t} \approx 1 + \beta t \qquad \text{and} \qquad e^{-\beta t} \approx 1 - \beta t$$

$$\therefore \quad y = e^{-bt}\left[C(1+\beta t) + D(1-\beta t)\right]$$

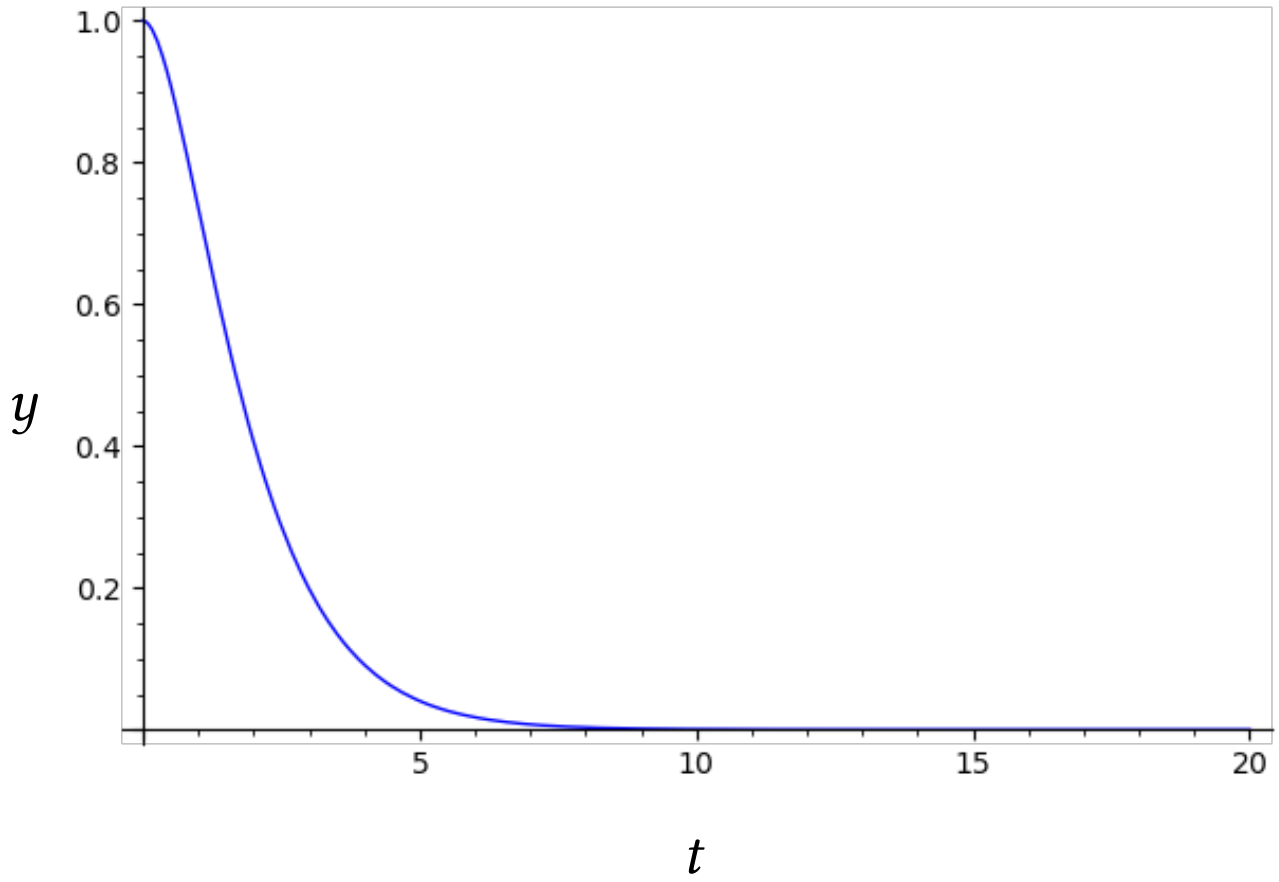
By rearranging,

$$y = e^{-bt}\left[(C+D) + (C-D)\beta t\right]$$

Calling $C + D = A_1$ and $C - D = A_2$

$$\boxed{y = e^{-bt}(A_1 + A_2\beta t)}$$

- As t is present in e^{-bt} and also in the term $A_2\beta t$, both of them contribute to the variation of y with respect to time.
- But by the virtue of having t in the exponent, the term e^{-bt} predominantly contributes to the equation.
- Only for small values of t, the term $A_2\beta t$ contributes in a magnitude comparable to that of e^{-bt}.
- Therefore, though y decreases throughout with increase of t, the decrement is slow in the beginning, and then decreases rapidly to approach the value zero.
- Such damping motion of a body is called *critical damping*.
- This type of motion is exhibited by many pointer instruments such as voltmeter, ammeter etc in which the pointer moves to the correct position and comes to rest without any oscillation.


1.0
0.8
0.6
0.4
0.2
y
5
10
15
20
t

Case 3: Low damping or under damping ($b^2 < \omega^2$)

In this case $\sqrt{b^2 - \omega^2}$ is imaginary. Let us write,

$$\sqrt{b^2 - \omega^2} = i\sqrt{\omega^2 - b^2} = i\beta' \quad \text{where } i = \sqrt{-1}$$

Then the equation (4) becomes:

$$\begin{aligned} y &= Ce^{(-b+i\beta')t} + De^{(-b-i\beta')t} \\ &= e^{-bt}\left(Ce^{i\beta' t} + De^{-i\beta' t}\right) \\ &= e^{-bt}\left\{C\left[\cos(\beta' t) + i\sin(\beta' t)\right] + D\left[\cos(\beta' t) - i\sin(\beta' t)\right]\right\} \\ &= e^{-bt}\left[(C+D)\cos(\beta' t) + i(C-D)\sin(\beta' t)\right] \end{aligned}$$

Rewriting $(C + D) = A \sin\phi$ and $i(C - D) = A \cos\phi$ where A and ϕ are constants,

$$y = e^{-bt} \left[A \sin\phi \cos(\beta' t) + A \cos\phi \sin(\beta' t)\right]$$

$$y = Ae^{-bt} \sin(\beta' t + \phi)$$

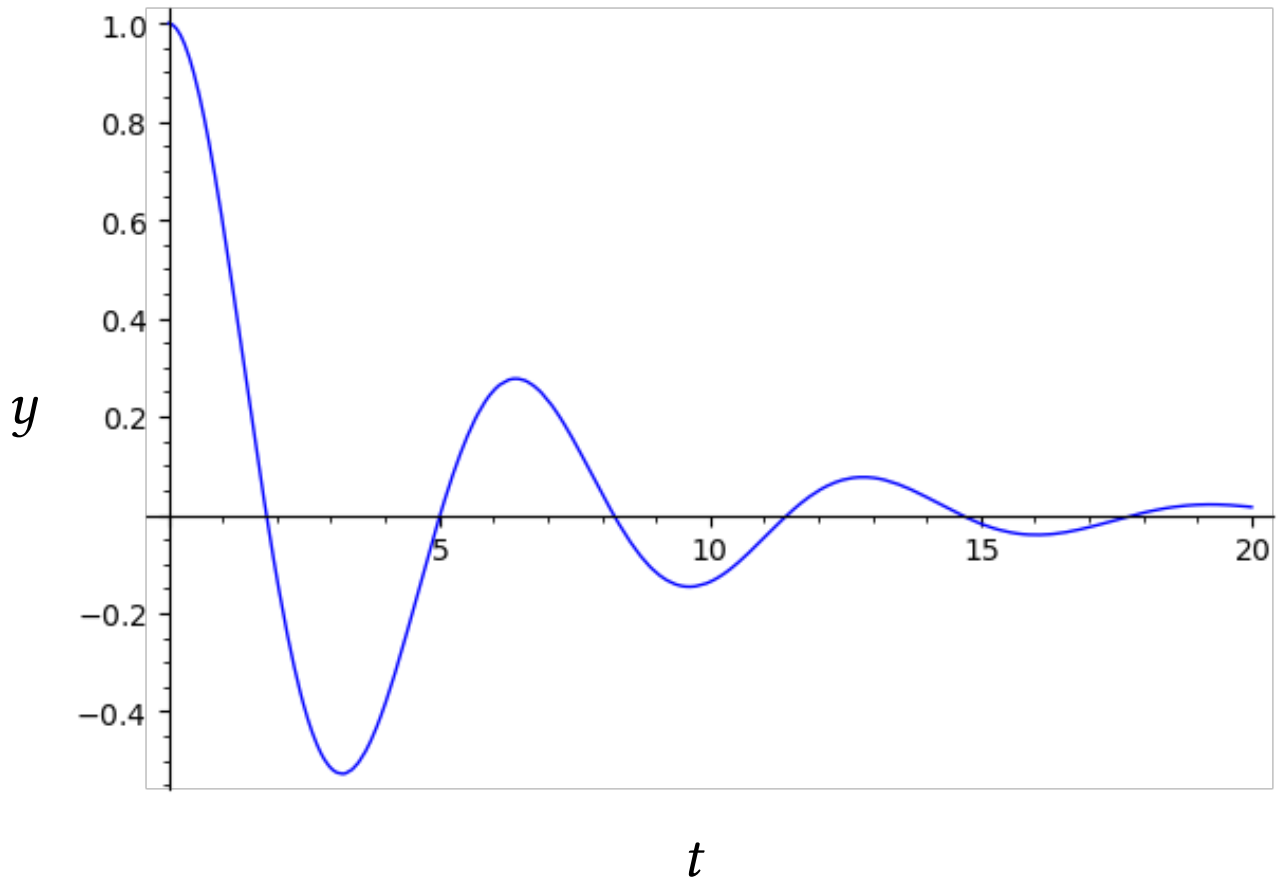
This equation represents displacement in the under damped harmonic oscillator. The oscillations are not simple harmonic because the amplitude Ae^{bt} is not constant and decreases with time t. The decay of amplitude depends upon the damping factor b.

Time period:

$$T = \frac{2\pi}{\beta'} = \frac{2\pi}{\sqrt{\omega^2 - b^2}} = \frac{2\pi}{\sqrt{\frac{k}{m} - \frac{r^2}{4m^2}}}$$

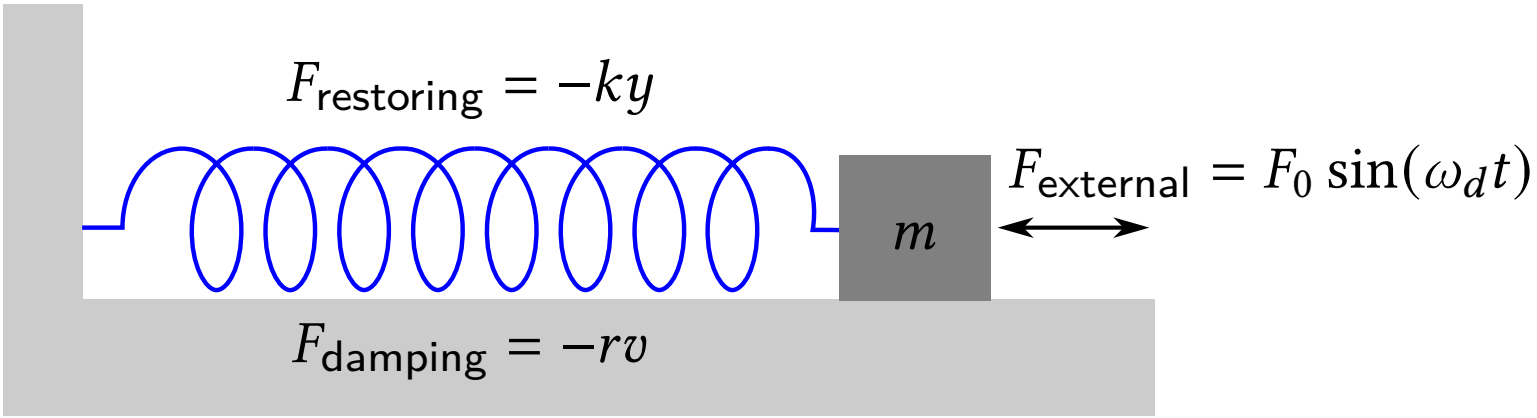
Frequency:

$$f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{m} - \frac{r^2}{4m^2}}$$


y
1.0
0.8
0.6
0.4
0.2
−0.2
−0.4
5
10
15
20
t

Forced Vibrations

- When a body is set in oscillation under an external periodic force whose frequency is not equal to the natural frequency of the body then in the beginning the body tries to oscillate with its own natural frequency.
- But very soon these oscillations die out and the body begins to oscillate with the frequency of the external periodic force (called the driving force).
- Such oscillations of the body are called the forced harmonic oscillations and the particle executing such vibrations is called the forced harmonic oscillator.
- The amplitude of forced oscillations depends upon the relation between the frequency of the applied force and the natural frequency of the body.



- Consider a body of mass m connected to a spring with spring constant k and subjected to a damping force with damping constant r.
- When it is displaced and released it starts oscillating about a mean position.
- If the body is driven by an external periodic force of the form $F_0 \sin(\omega_d t)$ then the oscillator is forced to oscillate at the angular frequency ω_d of this external periodic force.

At any instant the forces acting on the body are:

a) A restoring force proportional to the displacement but oppositely directed which is given by

$$F_{\text{restoring}} = -ky$$

where k is the spring constant or force constant.

b) A frictional or damping force proportional to velocity but oppositely directed which is given by

$$F_{\text{damping}} = -r\frac{dy}{dt}$$

where r is the frictional force per unit velocity, also called the damping constant.

c) An applied external periodic force given by

$$F_{\text{external}} = F_0 \sin(\omega_d t)$$

where F_0 is the amplitude and ω_d is the angular frequency of the driving force.

The total force acting on the particle is given by

$$F = F_{\text{restoring}} + F_{\text{damping}} + F_{\text{external}} = -ky - r\frac{dy}{dt} + F_0 \sin(\omega_d t)$$

From Newton's second law of motion

$$F = m\frac{d^2y}{dt^2}$$

Putting net force into this equation we get

$$m\frac{d^2y}{dt^2} = -ky - r\frac{dy}{dt} + F_0 \sin(\omega_d t)$$

By rearranging it we get

$$\frac{d^2y}{dt^2} + \frac{r}{m}\frac{dy}{dt} + \frac{k}{m}y = \frac{F_0}{m}\sin(\omega_d t) \tag{1}$$

Let

$$\frac{r}{m} = 2b, \qquad \frac{k}{m} = \omega^2, \qquad \frac{F_0}{m} = f \tag{2}$$

With this we can rewrite the differential equation as

$$\frac{d^2y}{dt^2} + 2b\frac{dy}{dt} + \omega^2 y = f\sin(\omega_d t) \tag{3}$$

Solution of this differential equation will be of the form

$$y = A\sin(\omega_d t - \phi) \tag{4}$$

Where A is the amplitude of the vibrations and ϕ is the phase difference.

To get A and ϕ differentiate equation (4) with respect to t

$$\frac{dy}{dt} = \omega_d A \cos(\omega_d t - \phi)$$

$$\frac{d^2y}{dt^2} = -\omega_d^2 A \sin(\omega_d t - \phi)$$

and substitute them in equation (3)

$$-\omega_d^2 A \sin(\omega_d t-\phi)+2b\omega_d A \cos(\omega_d t-\phi)+\omega^2 A \sin(\omega_d t-\phi) = f \sin(\omega_d t-\phi+\phi)$$

$$\implies \quad (\omega^2 - \omega_d^2)A \sin(\omega_d t - \phi) + 2b\omega_d A \cos(\omega_d t - \phi) = f \sin(\omega_d t - \phi) \cos\phi + f \cos(\omega_d t - \phi) \sin\phi$$

If this equation holds for all values of t, then the coefficients of $\sin(\omega_d t - \phi)$ and $\cos(\omega_d t - \phi)$ must be equal on both sides.

Or to say that the coefficients (the terms in the square brackets) are zero in the following rearranged form of the equation.

$$\left[(\omega^2 - \omega_d^2)A - f \cos\phi\right] \sin(\omega_d t - \phi) + \left[2b\omega_d A - f \sin\phi\right] \cos(\omega_d t - \phi) = 0$$

Therefore, on comparing we get these two relations

$$(\omega^2 - \omega_d^2)A = f\cos\phi \tag{5}$$

$$2b\omega_d A = f\sin\phi \tag{6}$$

By squaring and adding equations (5) and (6) we get

$$(\omega^2 - \omega_d^2)^2 A^2 + 4b^2\omega_d^2 A^2 = f^2$$

$$\therefore \quad A = \frac{f}{\sqrt{(\omega^2 - \omega_d^2)^2 + 4b^2\omega_d^2}} \tag{7}$$

By dividing equation (6) from (5) we get

$$\tan\phi = \frac{2b\omega_d}{\omega^2 - \omega_d^2} \qquad \Longrightarrow \quad \phi = \arctan\left(\frac{2b\omega_d}{\omega^2 - \omega_d^2}\right) \tag{8}$$

Hence, solution for the forced harmonic oscillator can be written as,

$$y = \frac{f}{\sqrt{(\omega^2 - \omega_d^2)^2 + 4b^2\omega_d^2}} \sin(\omega_d t - \phi) \tag{9}$$

From the above equations it is clear that the amplitude A and phase ϕ of the forced oscillations depend on the driving frequency ω_d and the natural frequency ω of the oscillator.

Case 1: When driving frequency is low $(\omega_d \ll \omega)$

We can approximate

$$\sqrt{(\omega^2 - \omega_d^2)^2 + 4b^2\omega_d^2} \approx \sqrt{(\omega^2)^2} = \omega^2$$

$$\therefore \quad A = \frac{f}{\sqrt{(\omega^2 - \omega_d^2)^2 + 4b^2\omega_d^2}} \approx \frac{f}{\omega^2} = \frac{F_0/m}{k/m} = \frac{F_0}{k}$$

and

$$\phi = \arctan\left(\frac{2b\omega_d}{\omega^2 - \omega_d^2}\right) \approx \arctan(0) = 0$$

It shows that the force and displacement are in phase.

Case 2: When driving frequency is equal to natural frequency $(\omega_d = \omega)$

This frequency is called *resonant frequency*.

$$\therefore \quad A = \frac{f}{\sqrt{(\omega^2 - \omega_d^2)^2 + 4b^2\omega_d^2}} = \frac{f}{\sqrt{4b^2\omega_d^2}} = \frac{f}{2b\omega_d} = \frac{F_0/m}{(r/m)\omega_d} = \frac{F_0}{r\omega_d}$$

The amplitude of the vibrations becomes *maximum* and is inversely proportional to the damping coefficient.

$$\phi = \arctan\left(\frac{2b\omega_d}{\omega^2 - \omega_d^2}\right) = \arctan\left(\frac{2b\omega_d}{0}\right) = \arctan(\infty) = \frac{\pi}{2}$$

The displacement lags behind the force by a phase of $\pi/2$.

Case 3: When driving frequency is high $(\omega_d \gg \omega)$

We can approximate

$$\sqrt{(\omega^2 - \omega_d^2)^2 + 4b^2\omega_d^2} \approx \sqrt{(-\omega_d^2)^2} = \omega_d^2$$

$$\therefore \quad A = \frac{f}{\sqrt{(\omega^2 - \omega_d^2)^2 + 4b^2\omega_d^2}} \approx \frac{f}{\omega_d^2} = \frac{F_0}{m\omega_d^2}$$

and

$$\phi = \arctan\left(\frac{2b\omega_d}{\omega^2 - \omega_d^2}\right) \approx \arctan\left(\frac{2b\omega_d}{-\omega_d^2}\right) = \arctan\left(\frac{2b}{-\omega_d}\right) \approx \arctan(0) = \pi$$

Because of negative sign in the argument of arctan it will be in second quadrant, therefore π. The displacement lags behind the force by a phase of π.

Resonance

Resonance is a phenomenon in which a body vibrates with maximum amplitude under the influence of an external vibration whose frequency is equal to the natural frequency of the vibrating body.

a) Condition for Amplitude Resonance

In case of forced vibrations, the expression for amplitude A and phase ϕ is given by,

$$A = \frac{f}{\sqrt{(\omega^2 - \omega_d^2)^2 + 4b^2\omega_d^2}} \qquad \text{and} \qquad \phi = \arctan\left(\frac{2b\omega_d}{\omega^2 - \omega_d^2}\right)$$

The amplitude expression shows its variation with the frequency of the driving force ω_d. For a particular value of ω_d, the amplitude becomes maximum. The phenomenon of amplitude becoming a maximum is known as *amplitude resonance*.

The amplitude is maximum when $\sqrt{(\omega^2 - \omega_d^2)^2 + 4b^2\omega_d^2}$ is minimum. If the damping b is small the maximum amplitude reduces to

$$A_{\text{max}} = \frac{f}{2b\omega_d}$$

b) Sharpness of Resonance

- We have seen that the amplitude of the forced oscillations is maximum when the frequency of the applied force is at resonant frequency.
- If the frequency changes from this resonance value, the amplitude falls. When the fall in amplitude for a small change from the resonance condition is very large, the resonance is said to be *sharp* and if the fall in amplitude is small, the resonance is termed as *flat*.
- *Thus the term sharpness of resonance can be defined as the rate of fall in amplitude with respect to the change in driving frequency on either side of the resonant frequency.*

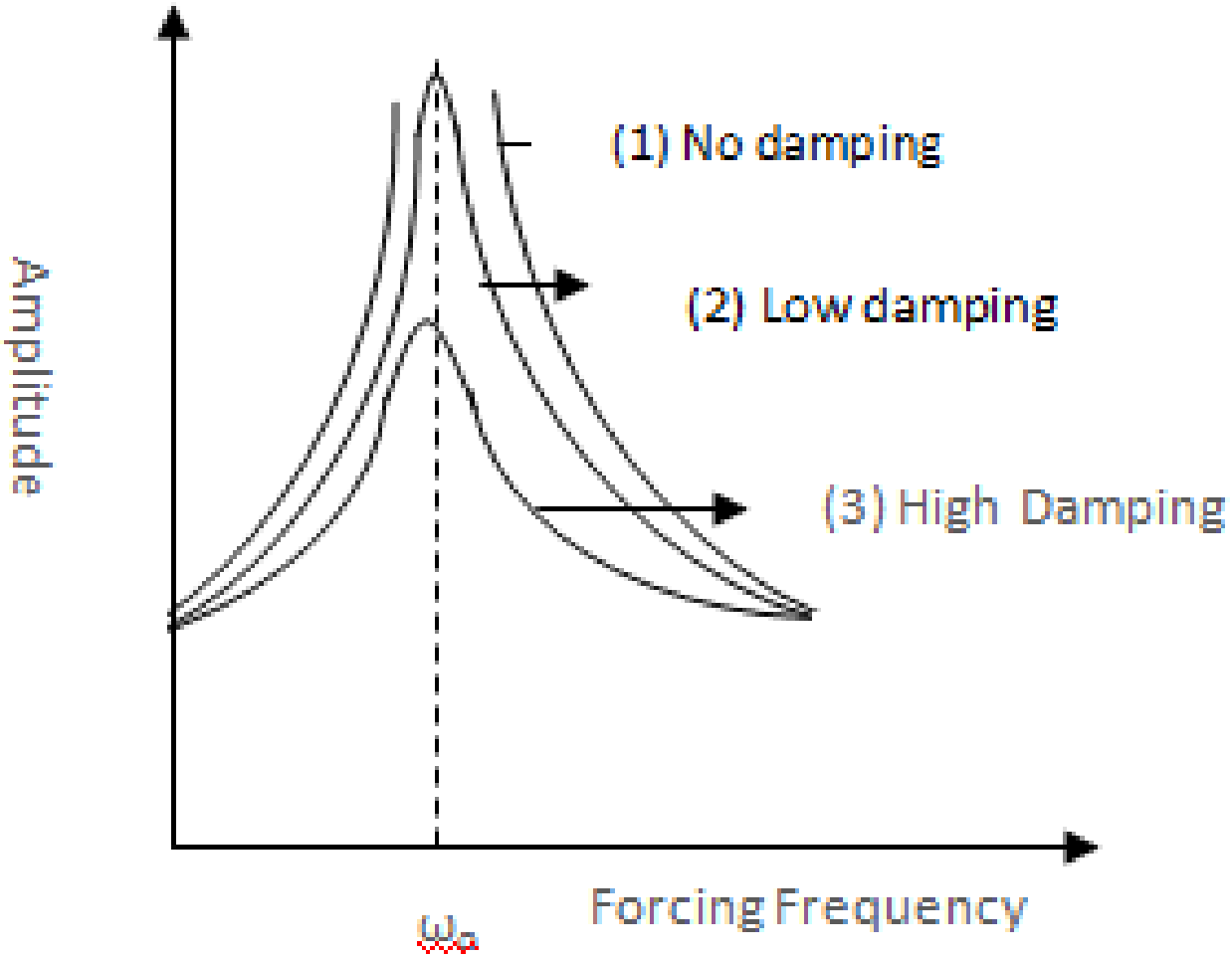


Figure: Effect of damping on sharpness of resonance.

- Figure shows the variation of amplitude with driving frequency for different amounts of damping.
- Curve (1) shows the variation of amplitude when there is no damping i.e., $b = 0$. In this case the amplitude is infinite at $\omega_d = \omega$. This case is never realized in practice due to presence of friction/dissipation forces always contributing to some damping factor.
- Curves (2) and (3) show the variation of amplitude with respect to low and high damping. It can be seen that the resonant peak moves towards the left as the damping factor is increased.
- It is also observed that the value of amplitude, which is different for different values of b (damping), diminishes as the value of b increases.
- This indicates that the smaller is the damping, sharper is the resonance and if large is the damping, flatter is the resonance.

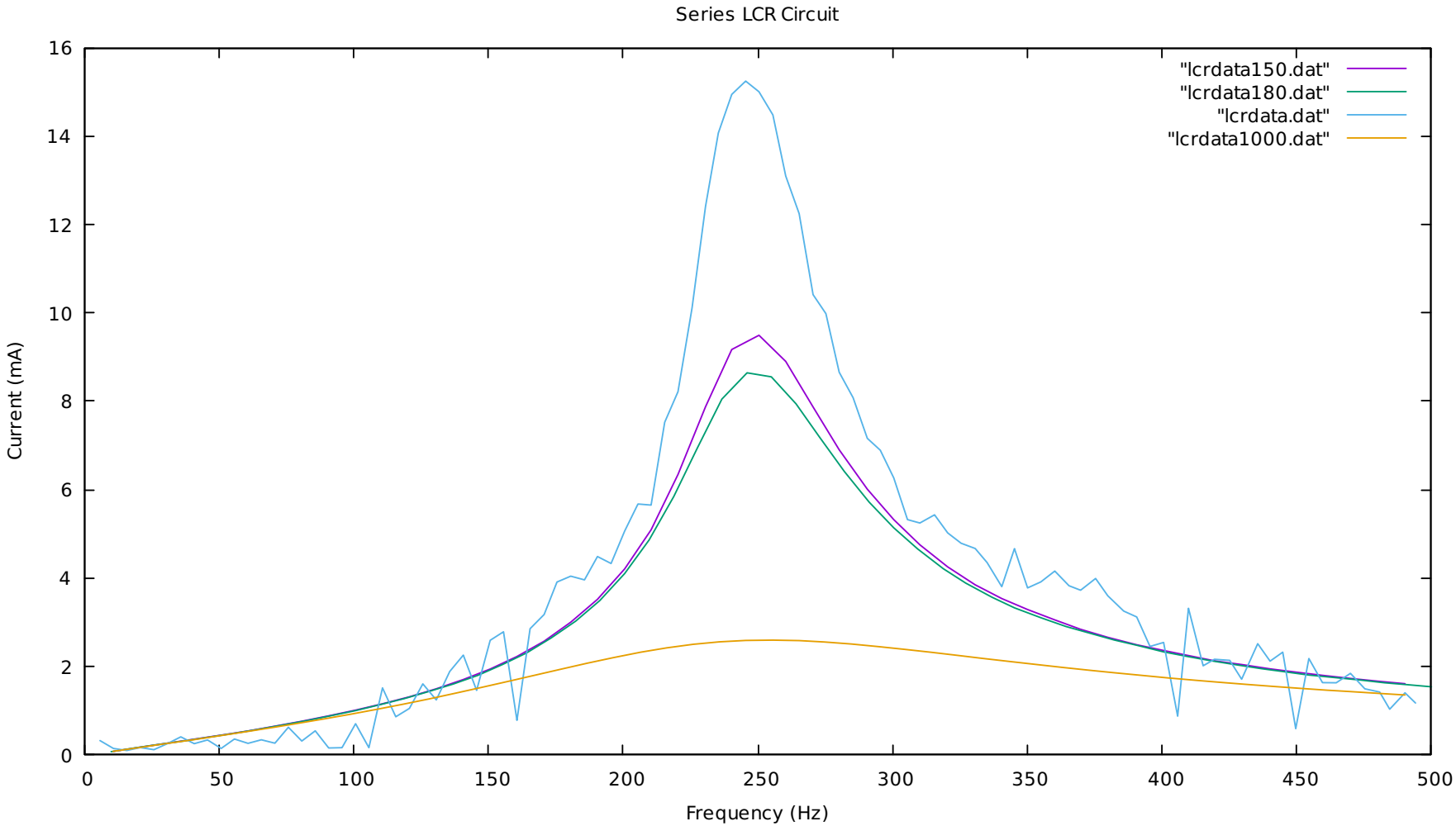


Figure: Effect of damping on sharpness of resonance.

References

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